ExamGOAL
PYQ PDF

Free · With answer key

JEE MAIN · QUESTION PAPER

JEE Main 2021 (Online) 17th March Morning Shift

JEE Main Question Paper with Answer Key

17 Mar 2021	90	20	English
HELD ON	QUESTIONS	OUT OF SYLLABUS	LANGUAGE

INSIDE THIS PDF

Chemistry	22 questions
Mathematics	20 questions
Physics	28 questions
Out of syllabus	20 questions
Answer key	at the end

Practise online

Every question with its full solution, free on **questions.examside.com**

1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.
3. **Understand**
   Scan the QR code beside any question for its full solution.

Chemistry 22 questions

Q1. MCQ (Single Correct Answer)

The absolute value of the electron gain enthalpy of halogens satisfies :

(A) Cl > Br > F > I (B) Cl > F > Br > I (C) I > Br > Cl > F (D) F > Cl > Br > I

Open solution

Q2. MCQ (Single Correct Answer)

What is the spin-only magnetic moment value (BM) of a divalent metal ion with atomic number 25, in it's aqueous solution?

(A) zero (B) 5.26 (C) 5.0 (D) 5.92

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

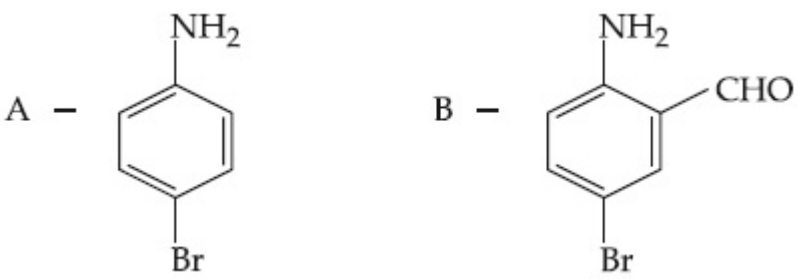
Q3. MCQ (Single Correct Answer)

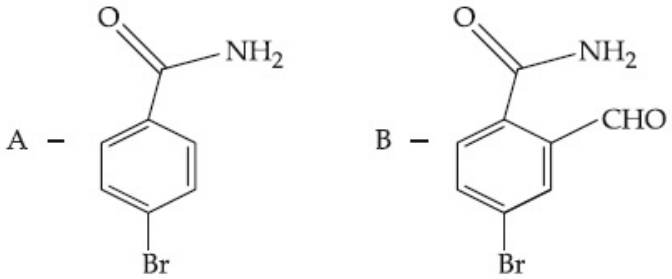
Open solution

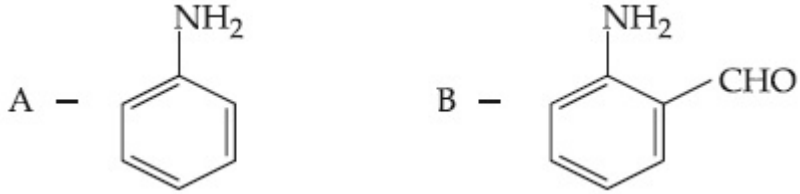
Hoffmann bromomide degradation of benzamide gives product A, which upon heating with $CHCl_3$ and NaOH gives product B.

The structures of A and B are :

(A) 

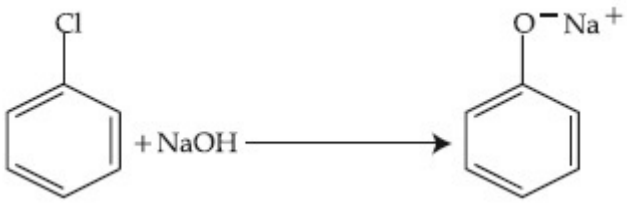
(B) 

(C) 

(D) 

Q4. MCQ (Single Correct Answer)

Open solution



The above reaction requires which of the following reaction conditions?

(A) 623 K, Cu, 300 atm

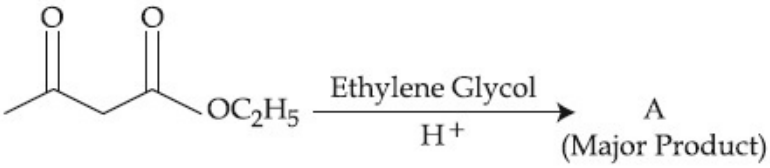
(B) 623 K, 300 atm

(C) 573 K, 300 atm

(D) 573 K, Cu, 300 atm

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

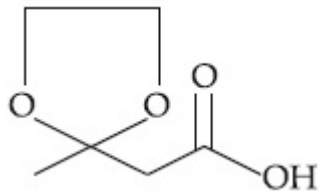
Q5. MCQ (Single Correct Answer)



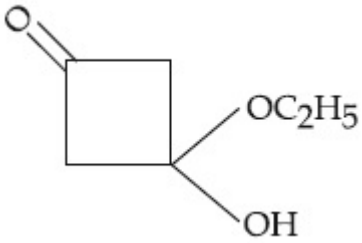
Open solution

The product 'A' in the above reaction is :

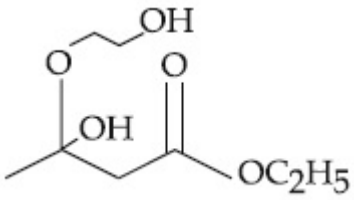
(A)



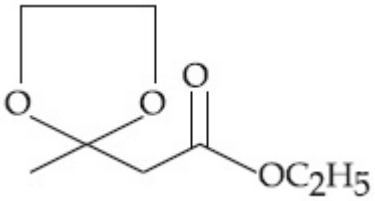
(B)



(C)

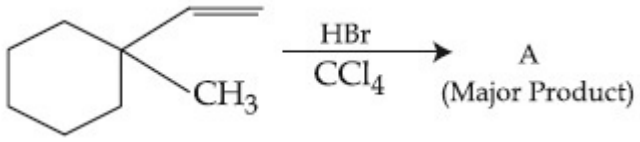


(D)



Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

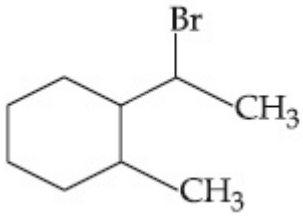
Q6. MCQ (Single Correct Answer)



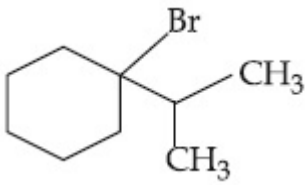
Open solution

Product 'A' in the above chemical reaction is :

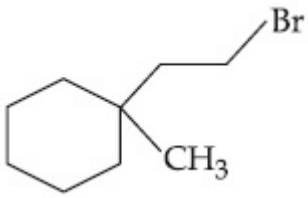
(A)



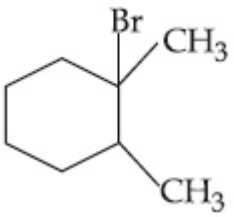
(B)



(C)



(D)



Q7. MCQ (Single Correct Answer)

Which of the following compound CANNOT act as a Lewis base?

(A) SF_4 (B) NF_3 (C) ClF_3 (D) PCl_5

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q8. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : Potassium permanganate on heating at 573K forms potassium manganate.

Statement II : Both potassium permanganate and potassium manganate are tetrahedral and paramagnetic in nature.

In the light of the above statements, choose the most appropriate answer from the options given below :

(A) Statement I is true but statement II is false

(B) Statement I is false but statement II is true

(C) Both statement I and statement II are false

(D) Both statement I and statement II are true

Q9. MCQ (Single Correct Answer)

Open solution

Mesityl oxide is a common name of :

(A) 3-Methyl cyclohexane carbaldehyde

(B) 4-Methyl pent-3-en-2-one

(C) 2,4-Dimethyl pentan-3-one

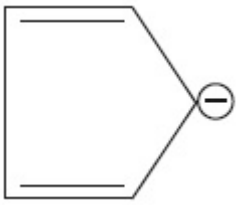
(D) 2-Methyl cyclohexanone

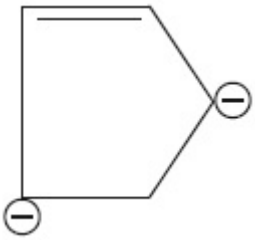
Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

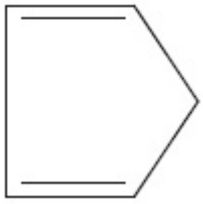
Q10. MCQ (Single Correct Answer)

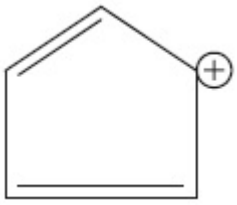
Open solution

Which of the following is an aromatic compound?

(A) 

(B) 

(C) 

(D) 

Q11. MCQ (Single Correct Answer)

Open solution

A central atom in a molecule has two lone pairs of electrons and forms three single bonds. The shape of this molecule is :

(A) trigonal pyramidal

(B) T-shaped

(C) see-saw

(D) planar triangular

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q12. MCQ (Single Correct Answer)

Open solution

Given below are two statements :

Statement I : Retardation factor (R_f) can be measured in meter/centimeter.

Statement II : R_f value of a compound remains constant in all solvents.

Choose the most appropriate answer from the options given below :

(A) Statement I is true but statement II is false

(B) Statement I is false but statement II is true

(C) Both statement I and statement II are true

(D) Both statement I and statement II are false

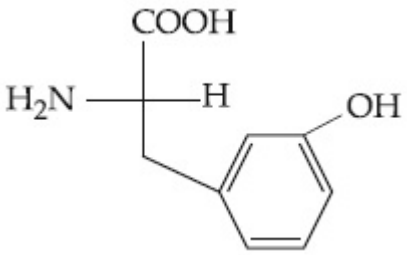
Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q13. MCQ (Single Correct Answer)

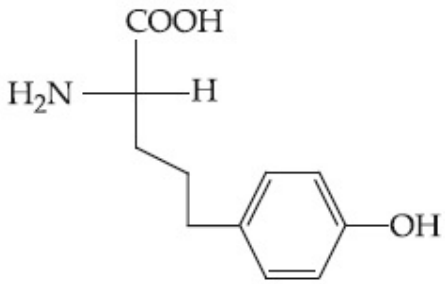
Open solution

Which of the following is correct structure of tyrosine?

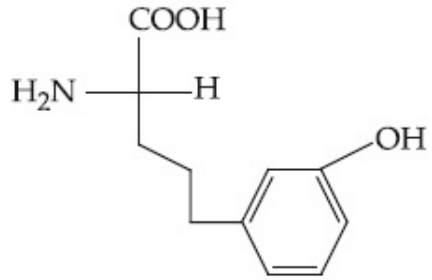
(A)



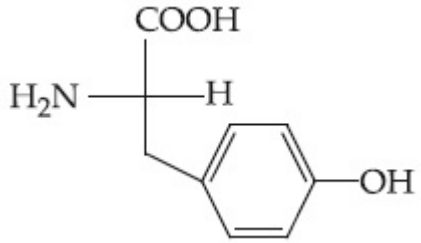
(B)



(C)



(D)



Q14. MCQ (Single Correct Answer)

Open solution

Which of the following reactions is an example of ammonolysis?

(A) $C_6H_5COCl + C_6H_5NH_2 \rightarrow C_6H_5CONHC_6H_5$

(B) $C_6H_5CH_2CN \xrightarrow{[H]} C_6H_5CH_2CH_2NH_2$

(C) $C_6H_5CH_2CN \xrightarrow{HCl} C_6H_5\overset{+}{N}H_3Cl^-$

(D) $C_6H_5CH_2Cl + NH_3 \rightarrow C_6H_5CH_2NH_2$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q15. Numerical

A certain orbital has n = 4 and $m_L = -3$. The number of radial nodes in this orbital is _____. (Round off to the Nearest Integer).

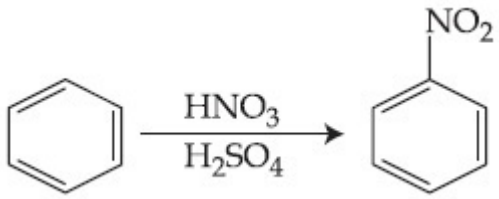
Open solution

Q16. Numerical

15 mL of aqueous solution of Fe^{2+} in acidic medium completely reacted with 20 mL of 0.03 M aqueous $Cr_2O_7^{2-}$. The molarity of the Fe^{2+} solution is _____ $\times 10^{-2}$ M. (Round off to the Nearest Integer).

Open solution

Q17. Numerical



Open solution

In the above reaction, 3.9 g of benzene on nitration gives 4.92 g of nitrobenzene. The percentage yield of nitrobenzene in the above reaction is _____%. (Round off to the Nearest Integer).

(Given atomic mass : C : 12.0 u, H : 1.0 u, O : 16.0 u, N : 14.0 u)

Q18. Numerical

For a certain first order reaction 32% of the reactant is left after 570s. The rate constant of this reaction is _____ $\times 10^{-3}$ s^{-1}. (Round off to the Nearest Integer). [Given : $\log_{10}2 = 0.301$, $\ln 10 = 2.303$]

Open solution

Q19. Numerical

0.01 moles of a weak acid HA ($K_a = 2.0 \times 10^{-6}$) is dissolved in 1.0 L of 0.1 M HCl solution. The degree of dissociation of HA is _____ $\times 10^{-5}$ (Round off to the Nearest Integer).

[Neglect volume change on adding HA. Assume degree of dissociation <<1]

Open solution

Q20. Numerical

The mole fraction of a solute in a 100 molal aqueous solution is _____ $\times 10^{-2}$. (Round off to the Nearest Integer).

[Given : Atomic masses : H : 1.0 u, O : 16.0 u]

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q21. Numerical

The standard enthalpies of formation of Al_2O_3 and CaO are -1675 kJ mol^{-1} and -635 kJ mol^{-1} respectively.

For the reaction

$3CaO + 2Al \rightarrow 3Ca + Al_2O_3$ the standard reaction enthalpy Δ_rH^0 = _____ kJ.

(Round off to the Nearest Integer)

Open solution

Q22. Numerical

The oxygen dissolved in water exerts a partial pressure of 20 kPa in the vapour above water. The molar solubility of oxygen in water is _____ $\times 10^{-5}$ mol dm^{-3}. (Round off to the Nearest Integer).

[Given : Henry's law constant = $K_H = 8.0 \times 10^4$ kPa for O_2. Density of water with dissolved oxygen = 1.0 kg dm^{-3}]

Open solution

Mathematics 20 questions

Q23. MCQ (Single Correct Answer)

In a triangle PQR, the co-ordinates of the points P and Q are $(-2, 4)$ and $(4, -2)$ respectively. If the equation of the perpendicular bisector of PR is $2x - y + 2 = 0$, then the centre of the circumcircle of the ΔPQR is :

(A) $(-1, 0)$ (B) $(1, 4)$ (C) $(0, 2)$ (D) $(-2, -2)$

Open solution

Q24. MCQ (Single Correct Answer)

The value of $\lim\limits_{x\to 0^+} \frac{\cos^{-1}(x-[x]^2).\sin^{-1}(x-[x]^2)}{x-x^3}$, where [x] denotes the greatest integer $\leq$ x is :

(A) π (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{2}$ (D) 0

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q25. MCQ (Single Correct Answer)

Which of the following statements is correct for the function g(α) for $\alpha \in$ R such that

$$g(\alpha) = \int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sin^{\alpha} x}{\cos^{\alpha} x + \sin^{\alpha} x} dx$$

Open solution

(A) $g(\alpha)$ is a strictly increasing function

(B) $g(\alpha)$ is an even function

(C) $g(\alpha)$ has an inflection point at $\alpha = -\frac{1}{2}$

(D) $g(\alpha)$ is a strictly decreasing function

Q26. MCQ (Single Correct Answer)

Which of the following is true for y(x) that satisfies the differential equation

$\frac{dy}{dx}$ = xy $-$ 1 + x $-$ y; y(0) = 0 :

Open solution

(A) y(1) = 1

(B) $y(1) = e^{-\frac{1}{2}} - 1$

(C) $y(1) = e^{\frac{1}{2}} - e^{-\frac{1}{2}}$

(D) $y(1) = e^{\frac{1}{2}} - 1$

Q27. MCQ (Single Correct Answer)

The sum of possible values of x for

$\tan^{-1}(x + 1) + \cot^{-1}\left(\frac{1}{x-1}\right) = \tan^{-1}\left(\frac{8}{31}\right)$ is :

Open solution

(A) $-\frac{32}{4}$

(B) $-\frac{33}{4}$

(C) $-\frac{31}{4}$

(D) $-\frac{30}{4}$

Q28. MCQ (Single Correct Answer)

The value of $4 + \frac{1}{5+\frac{1}{4+\frac{1}{5+\frac{1}{4+.....\infty}}}}$ is :

Open solution

(A) $2 + \frac{2}{5}\sqrt{30}$

(B) $2 + \frac{4}{\sqrt{5}}\sqrt{30}$

(C) $5 + \frac{2}{5}\sqrt{30}$

(D) $4 + \frac{4}{\sqrt{5}}\sqrt{30}$

Q29. MCQ (Single Correct Answer)

The inverse of $y = 5^{\log x}$ is :

Open solution

(A) $x = 5^{\log y}$

(B) $x = y^{\frac{1}{\log 5}}$

(C) $x = 5^{\frac{1}{\log y}}$

(D) $x = y^{\log 5}$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q30. MCQ (Single Correct Answer)

Two dies are rolled. If both dices have six faces numbered 1, 2, 3, 5, 7 and 11, then the probability that the sum of the numbers on the top faces is less than or equal to 8 is :

(A) $\frac{4}{9}$ (B) $\frac{1}{2}$ (C) $\frac{5}{12}$ (D) $\frac{17}{36}$

Open solution

Q31. MCQ (Single Correct Answer)

If the fourth term in the expansion of $(x + x^{\log_2 x})^7$ is 4480, then the value of x where x∈N is equal to :

(A) 3 (B) 1 (C) 4 (D) 2

Open solution

Q32. MCQ (Single Correct Answer)

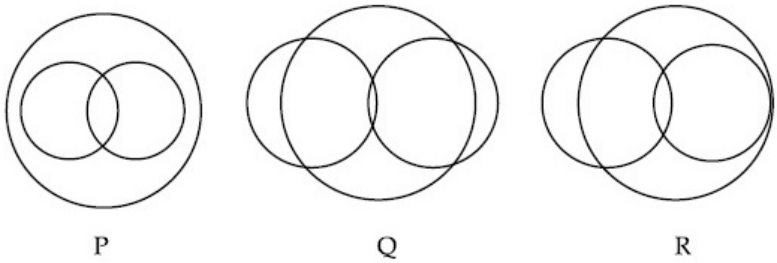
The system of equations kx + y + z = 1, x + ky + z = k and x + y + zk = k^2 has no solution if k is equal to :

(A) 0 (B) -1 (C) -2 (D) 1

Open solution

Q33. MCQ (Single Correct Answer)

In a school, there are three types of games to be played. Some of the students play two types of games, but none play all the three games. Which Venn diagrams can justify the above statement?



(A) Q and R (B) None of these (C) P and R (D) P and Q

Open solution

Q34. MCQ (Single Correct Answer)

If $\cot^{-1}(\alpha) = \cot^{-1} 2 + \cot^{-1} 8 + \cot^{-1} 18 + \cot^{-1} 32 +$ ..... upto 100 terms, then α is :

(A) 1.02 (B) 1.03 (C) 1.01 (D) 1.00

Open solution

Q35. MCQ (Single Correct Answer)

Team 'A' consists of 7 boys and n girls and Team 'B' has 4 boys and 6 girls. If a total of 52 single matches can be arranged between these two teams when a boy plays against a boy and a girl plays against a girl, then n is equal to :

(A) 5 (B) 2 (C) 4 (D) 6

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q36. MCQ (Single Correct Answer)

The area of the triangle with vertices A(z), B(iz) and C(z + iz) is :

(A) 1

(B) $\frac{1}{2}|z|^2$

(C) $\frac{1}{2}|z + iz|^2$

(D) $\frac{1}{2}$

Open solution

Q37. Numerical

If $(2021)^{3762}$ is divided by 17, then the remainder is _____.

Open solution

Q38. Numerical

Let there be three independent events E_1, E_2 and E_3. The probability that only E_1 occurs is α, only E_2 occurs is β and only E_3 occurs is γ. Let 'p' denote the probability of none of events occurs that satisfies the equations
$(\alpha - 2\beta)p = \alpha\beta$ and $(\beta - 3\gamma)p = 2\beta\gamma$. All the given probabilities are assumed to lie in the interval (0, 1).

Then, $\frac{Probability\ of\ occurrence\ of\ E_1}{Probability\ of\ occurrence\ of\ E_3}$ is equal to _____.

Open solution

Q39. Numerical

The maximum value of z in the following equation $z = 6xy + y^2$, where $3x + 4y \leq 100$ and $4x + 3y \leq 75$ for $x \geq 0$ and $y \geq 0$ is _____.

Open solution

Q40. Numerical

If [.] represents the greatest integer function, then the value of

$\left|\int_0^{\sqrt{\frac{\pi}{2}}} \left[[x^2] - \cos x\right] dx\right|$ is _____.

Open solution

Q41. Numerical

If $f(x) = \sin\left(\cos^{-1}\left(\frac{1-2^{2x}}{1+2^{2x}}\right)\right)$ and its first derivative with respect to x is $-\frac{b}{a}\log_e 2$ when x = 1, where a and b are integers, then the minimum value of $|a^2 - b^2|$ is _____ .

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q42. Numerical

If the function $f(x) = \frac{\cos(\sin x) - \cos x}{x^4}$ is continuous at each point in its domain and $f(0) = \frac{1}{k}$, then k is _____.

Open solution

Physics 28 questions

Q43. MCQ (Single Correct Answer)

A boy is rolling a 0.5 kg ball on the frictionless floor with the speed of 20 ms^{-1}. The ball gets deflected by an obstacle on the way. After deflection it moves with 5% of its initial kinetic energy. What is the speed of the ball now?

(A) 14.41 ms^{-1} (B) 19.0 ms^{-1} (C) 4.47 ms^{-1} (D) 1.00 ms^{-1}

Open solution

Q44. MCQ (Single Correct Answer)

A polyatomic ideal gas has 24 vibrational modes. What is the value of γ?

(A) 1.37 (B) 1.30 (C) 1.03 (D) 10.3

Open solution

Q45. MCQ (Single Correct Answer)

Two ideal polyatomic gases at temperatures T_1 and T_2 are mixed so that there is no loss of energy. If F_1 and F_2, m_1 and m_2, n_1 and n_2 be the degrees of freedom, masses, number of molecules of the first and second gas respectively, the temperature of mixture of these two gases is :

(A) $\frac{n_1F_1T_1+n_2F_2T_2}{F_1+F_2}$ (B) $\frac{n_1F_1T_1+n_2F_2T_2}{n_1F_1+n_2F_2}$

(C) $\frac{n_1T_1+n_2T_2}{n_1+n_2}$ (D) $\frac{n_1F_1T_1+n_2F_2T_2}{n_1+n_2}$

Open solution

Q46. MCQ (Single Correct Answer)

If an electron is moving in the nth orbit of the hydrogen atom, then its velocity (v_n) for the nth orbit is given as :

(A) $v_n \propto \frac{1}{n}$ (B) $v_n \propto n^2$ (C) $v_n \propto n$ (D) $v_n \propto \frac{1}{n^2}$

Open solution

Q47. MCQ (Single Correct Answer)

Two identical metal wires of thermal conductivities K_1 and K_2 respectively are connected in series. The effective thermal conductivity of the combination is :

(A) $\frac{2K_1K_2}{K_1+K_2}$ (B) $\frac{K_1+K_2}{K_1K_2}$

(C) $\frac{K_1+K_2}{2K_1K_2}$ (D) $\frac{K_1K_2}{K_1+K_2}$

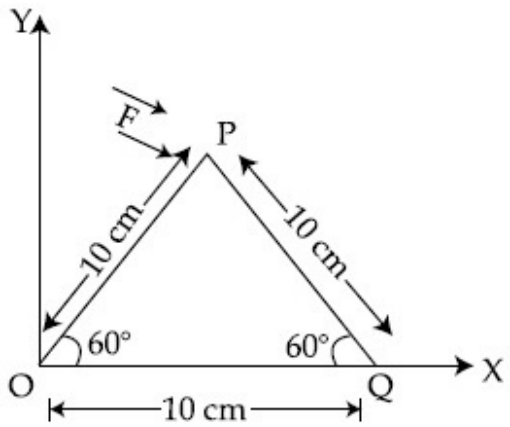
Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q48. MCQ (Single Correct Answer)

A triangular plate is shown. A force $\vec{F} = 4\hat{i} - 3\hat{j}$ is applied at point P. The torque at point P with respect to point 'O' and 'Q' are :

Open solution



(A) $-15 + 20\sqrt{3}, 15 + 20\sqrt{3}$

(B) $15 - 20\sqrt{3}, 15 + 20\sqrt{3}$

(C) $15 + 20\sqrt{3}, 15 - 20\sqrt{3}$

(D) $-15 - 20\sqrt{3}, 15 - 20\sqrt{3}$

Q49. MCQ (Single Correct Answer)

When two soap bubbles of radii a and b (b > a) coalesce, the radius of curvature of common surface is :

Open solution

(A) $\frac{b-a}{ab}$

(B) $\frac{a+b}{ab}$

(C) $\frac{ab}{a+b}$

(D) $\frac{ab}{b-a}$

Q50. MCQ (Single Correct Answer)

A car accelerates from rest at a constant rate α for some time after which it decelerates at a constant rate β to come to rest. If the total time elapsed is t seconds, the total distance travelled is :

Open solution

(A) $\frac{4\alpha\beta}{(\alpha+\beta)}t^2$

(B) $\frac{2\alpha\beta}{(\alpha+\beta)}t^2$

(C) $\frac{\alpha\beta}{2(\alpha+\beta)}t^2$

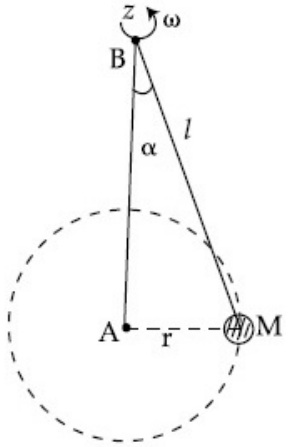
(D) $\frac{\alpha\beta}{4(\alpha+\beta)}t^2$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q51. MCQ (Single Correct Answer)

Open solution

A mass M hangs on a massless rod of length l which rotates at a constant angular frequency. The mass M moves with steady speed in a circular path of constant radius. Assume that the system is in steady circular motion with constant angular velocity ω. The angular momentum of M about point A is L_A which lies in the positive z direction and the angular momentum of M about point B is L_B. The correct statement for this system is :



(A) L_A is constant, both in magnitude and direction

(B) L_B is constant in direction with varying magnitude

(C) L_B is constant, both in magnitude and direction

(D) L_A and L_B are both constant in magnitude and direction

Q52. MCQ (Single Correct Answer)

Open solution

A solenoid of 1000 turns per metre has a core with relative permeability 500. Insulated windings of the solenoid carry an electric current of 5A. The magnetic flux density produced by the solenoid is : (permeability of free space = $4\pi \times 10^{-7}$ H/m)

(A) πT (B) $2 \times 10^{-3}\pi$ T (C) $10^{-4}\pi$ T (D) $\frac{\pi}{5}$ T

Q53. MCQ (Single Correct Answer)

Open solution

The thickness at the centre of a plano-convex lens is 3 mm and the diameter is 6 cm. If the speed of light in the material of the lens is 2×10^8 ms^{-1}, the focal length of the lens is _____.

(A) 0.30 cm (B) 30 cm (C) 15 cm (D) 1.5 cm

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q54. MCQ (Single Correct Answer)

For what value of displacement the kinetic energy and potential energy of a simple harmonic oscillation become equal ?

Open solution

(A) x = $\frac{A}{2}$

(B) x = $\pm$ A

(C) x = $\pm \frac{A}{\sqrt{2}}$

(D) x = 0

Q55. MCQ (Single Correct Answer)

An AC current is given by I = I_1 sinωt + I_2 cosωt. A hot wire ammeter will give a reading :

Open solution

(A) $\frac{I_1+I_2}{\sqrt{2}}$

(B) $\sqrt{\frac{I_1^2-I_2^2}{2}}$

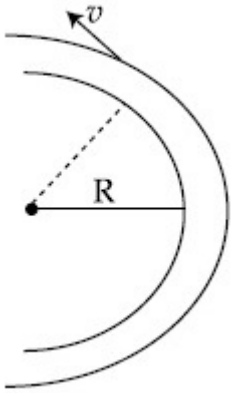
(C) $\sqrt{\frac{I_1^2+I_2^2}{2}}$

(D) $\frac{I_1+I_2}{2\sqrt{2}}$

Q56. MCQ (Single Correct Answer)

A modern grand - prix racing car of mass m is travelling on a flat track in a circular arc of radius R with a speed v. If the coefficient of static friction between the tyres and the track is μ_s, then the magnitude of negative lift F_L acting downwards on the car is : (Assume forces on the four tyres are identical and g = acceleration due to gravity)

Open solution



(A) $m\left(g-\frac{v^2}{\mu_s R}\right)$

(B) $-m\left(g+\frac{v^2}{\mu_s R}\right)$

(C) $m\left(\frac{v^2}{\mu_s R}-g\right)$

(D) $m\left(\frac{v^2}{\mu_s R}+g\right)$

Q57. MCQ (Single Correct Answer)

The vernier scale used for measurement has a positive zero error of 0.2 mm. If while taking a measurement it was noted that '0' on the vernier scale lies between 8.5 cm and 8.6 cm, vernier coincidence is 6, then the correct value of measurement is _____ cm. (least count = 0.01 cm)

Open solution

(A) 8.58 cm

(B) 8.54 cm

(C) 8.56 cm

(D) 8.36 cm

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q58. MCQ (Single Correct Answer)

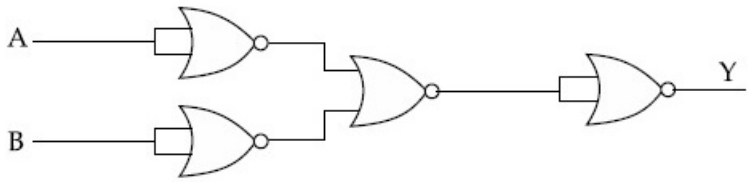
A current of 10A exists in a wire of cross-sectional area of 5 mm^2 with a drift velocity of 2×10^{-3} ms^{-1}. The number of free electrons in each cubic meter of the wire is _____.

(A) 625×10^{25} (B) 1×10^{23} (C) 2×10^{25} (D) 2×10^{6}

Open solution

Q59. MCQ (Single Correct Answer)

The output of the given combination gates represents :



(A) AND Gate (B) NOR Gate (C) NAND Gate (D) XOR Gate

Open solution

Q60. MCQ (Single Correct Answer)

Which level of the single ionized carbon has the same energy as the ground state energy of hydrogen atom?

(A) 8 (B) 6 (C) 1 (D) 4

Open solution

Q61. MCQ (Single Correct Answer)

An electron of mass m and a photon have same energy E. The ratio of wavelength of electron to that of photon is : (c being the velocity of light)

(A) $\frac{1}{c}\left(\frac{2m}{E}\right)^{1/2}$ (B) $\frac{1}{c}\left(\frac{E}{2m}\right)^{1/2}$

(C) $\left(\frac{E}{2m}\right)^{1/2}$ (D) $c(2mE)^{1/2}$

Open solution

Q62. Numerical

The radius in kilometer to which the present radius of earth (R = 6400 km) to be compressed so that the escape velocity is increased 10 times is _____.

Open solution

Q63. Numerical

The angular speed of truck wheel is increased from 900 rpm to 2460 rpm in 26 seconds. The number of revolutions by the truck engine during this time is _____. (Assuming the acceleration to be uniform).

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q64. Numerical

A parallel plate capacitor whose capacitance C is 14 pF is charged by a battery to a potential difference V = 12 V between its plates. The charging battery is now disconnected and a porcelain plate with k = 7 is inserted between the plates, then the plate would oscillate back and forth between the plates with a constant mechanical energy of _____ pJ. (Assume no friction)

Open solution

Q65. Numerical

The following bodies,

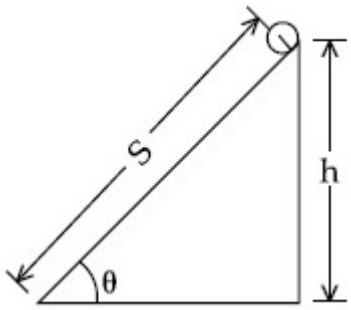
(1) a ring

(2) a disc

(3) a solid cylinder

(4) a solid sphere,

of same mass 'm' and radius 'R' are allowed to roll down without slipping simultaneously from the top of the inclined plane. The body which will reach first at the bottom of the inclined plane is _____. [Mark the body as per their respective numbering given in the question]



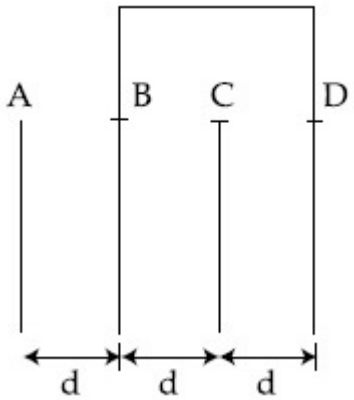
Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q66. Numerical

Four identical rectangular plates with length, l = 2 cm and breadth, b = $\frac{3}{2}$ cm are arranged as shown in figure. The equivalent capacitance between A and C is $\frac{x\varepsilon_0}{d}$. The value of x is _____.
(Round off to the Nearest Integer)

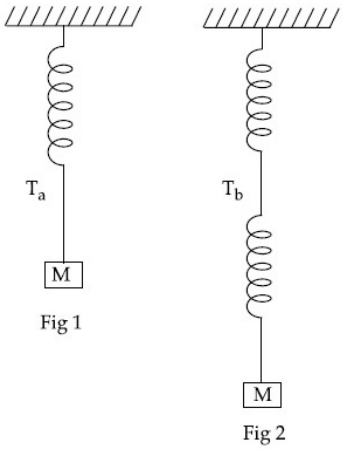
Open solution



Q67. Numerical

Consider two identical springs each of spring constant k and negligible mass compared to the mass M as shown. Fig. 1 shows one of them and Fig. 2 shows their series combination. The ratios of time period of oscillation of the two SHM is $T_b/T_a = \sqrt{x}$, where value of x is _____. (Round off to the Nearest Integer)

Open solution



Q68. Numerical

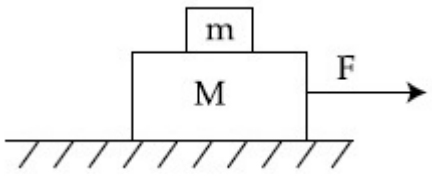
The equivalent resistance of series combination of two resistors is 's'. When they are connected in parallel, the equivalent resistance is 'p'. If s = np, then the minimum value for n is _____.
(Round off to the Nearest Integer)

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q69. Numerical

Two blocks (m = 0.5 kg and M = 4.5 kg) are arranged on a horizontal frictionless table as shown in figure. The coefficient of static friction between the two blocks is $\frac{3}{7}$. Then the maximum horizontal force that can be applied on the larger block so that the blocks move together is _____ N. (Round off to the Nearest Integer) [Take g as 9.8 ms^{-2}]



Open solution

Q70. Numerical

If 2.5×10^{-6} N average force is exerted by a light wave on a non-reflecting surface of 30 cm^2 area during 40 minutes of time span, the energy flux of light just before it falls on the surface is _____ W/cm^2. (Round off to the Nearest Integer)

(Assume complete absorption and normal incidence conditions are there)

Open solution

Out of syllabus 20 questions

These questions are on topics no longer in the current syllabus. They are kept for practice, after everything else.

Q71. MCQ (Single Correct Answer) · **Chemistry**

With respect to drug-enzyme interaction, identify the wrong statement.

(A) Allosteric inhibitor changes the enzyme's active site

(B) Allosteric inhibitor competes with the enzyme's active site

(C) Competitive inhibitor binds to the enzyme's active site

(D) Non-Competitive inhibitor binds to the allosteric site

Open solution

Q72. MCQ (Single Correct Answer) · **Chemistry**

A colloidal system consisting of a gas dispersed in a solid is called a/an :

(A) foam (B) aerosol (C) solid sol (D) gel

Open solution

Q73. MCQ (Single Correct Answer) · **Chemistry**

Reducing smog is a mixture of :

(A) Smoke, fog and O_3

(B) Smoke, fog and N_2O_3

(C) Smoke, fog and $CH_2 = CH - CHO$

(D) Smoke, fog and SO_2

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q74. MCQ (Single Correct Answer) · **Chemistry**

Open solution

The correct order of conductivity of ions in water is :

(A) $K^+ > Na^+ > Cs^+ > Rb^+$

(B) $Cs^+ > Rb^+ > K^+ > Na^+$

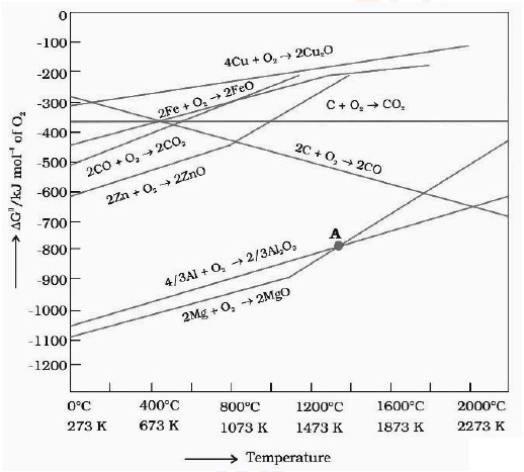
(C) $Na^+ > K^+ > Rb^+ > Cs^+$

(D) $Rb^+ > Na^+ > K^+ > Li^+$

Q75. MCQ (Single Correct Answer) · **Chemistry**

Open solution

The point of intersection and sudden increase in the slope, in the diagram given below, respectively, indicates :



(A) $\Delta G > 0$ and decomposition of the metal oxide

(B) $\Delta G < 0$ and decomposition of the metal oxide

(C) $\Delta G = 0$ and reduction of the metal oxide

(D) $\Delta G = 0$ and melting or boiling point of the metal oxide

Q76. MCQ (Single Correct Answer) · **Chemistry**

Open solution

The INCORRECT statement(s) about heavy water is (are)

(A) used as a moderator in nuclear reactor

(B) obtained as a by-product in fertilizer industry

(C) used for the study of reaction mechanism

(D) has a higher dielectric constant than water

Choose the correct answer from the options given below :

(A) (D) only (B) (B) and (D) only (C) (B) only (D) (C) only

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q77. Numerical · **Chemistry**

The reaction of white phosphorus on boiling with alkali in inert atmosphere resulted in the formation of product 'A'. The reaction of 1 mol of 'A' with excess of $AgNO_3$ in aqueous medium gives _____ mol(s) of Ag. (Round off to the Nearest Integer).

Open solution

Q78. Numerical · **Chemistry**

The pressure exerted by a non-reactive gaseous mixture of 6.4g of methane and 8.8 g of carbon dioxide in a 10L vessel at 27°C is _____ kPa. (Round off to the Nearest Integer).

[Assume gases are ideal, R = 8.314 J mol^{-1} K^{-1}

Atomic masses : C : 12.0 u, H : 1.0 u, O : 16.0 u]

Open solution

Q79. MCQ (Single Correct Answer) · **Mathematics**

Let $\vec{a} = 2\hat{i} - 3\hat{j} + 4\hat{k}$ and $\vec{b} = 7\hat{i} + \hat{j} - 6\hat{k}$.

If $\vec{r} \times \vec{a} = \vec{r} \times \vec{b}, \vec{r} \cdot (\hat{i} + 2\hat{j} + \hat{k}) = -3$, then $\vec{r} \cdot (2\hat{i} - 3\hat{j} + \hat{k})$ is equal to :

(A) 10 (B) 8 (C) 13 (D) 12

Open solution

Q80. MCQ (Single Correct Answer) · **Mathematics**

The line $2x - y + 1 = 0$ is a tangent to the circle at the point (2, 5) and the centre of the circle lies on $x - 2y = 4$. Then, the radius of the circle is :

(A) $5\sqrt{3}$ (B) $4\sqrt{5}$ (C) $3\sqrt{5}$ (D) $5\sqrt{4}$

Open solution

Q81. MCQ (Single Correct Answer) · **Mathematics**

If the Boolean expression $(p \Rightarrow q) \Leftrightarrow (q * (\sim p)$ is a tautology, then the boolean expression $(p * (\sim q))$ is equivalent to :

(A) $q \Rightarrow p$ (B) $p \Rightarrow q$ (C) $p \Rightarrow \sim q$ (D) $\sim q \Rightarrow p$

Open solution

Q82. MCQ (Single Correct Answer) · **Mathematics**

The equation of the plane which contains the y-axis and passes through the point (1, 2, 3) is :

(A) x + 3z = 0 (B) 3x − z = 0 (C) x + 3z = 10 (D) 3x + z = 6

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q83. MCQ (Single Correct Answer) · **Mathematics**

Choose the incorrect statement about the two circles whose equations are given below :

$x^2 + y^2 - 10x - 10y + 41 = 0$ and

$x^2 + y^2 - 16x - 10y + 80 = 0$

(A) Distance between two centres is the average of radii of both the circles.

(B) Both circles pass through the centre of each other.

(C) Circles have two intersection points.

(D) Both circle's centers lie inside region of one another.

Open solution

Q84. MCQ (Single Correct Answer) · **Mathematics**

If $A = \begin{pmatrix} 0 & \sin\alpha \\ \sin\alpha & 0 \end{pmatrix}$ and $\det\left(A^2 - \frac{1}{2}I\right) = 0$, then a possible value of α is :

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{6}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{3}$

Open solution

Q85. Numerical · **Mathematics**

If the equation of the plane passing through the line of intersection of the planes $2x - 7y + 4z - 3 = 0$, $3x - 5y + 4z + 11 = 0$ and the point $(-2, 1, 3)$ is $ax + by + cz - 7 = 0$, then the value of $2a + b + c - 7$ is _____.

Open solution

Q86. Numerical · **Mathematics**

If $A = \begin{bmatrix} 2 & 3 \\ 0 & -1 \end{bmatrix}$, then the value of $\det(A^4) + \det(A^{10} - (\text{Adj}(2A))^{10})$ is equal to _____.

Open solution

Q87. Numerical · **Mathematics**

If $\vec{a} = \alpha\hat{i} + \beta\hat{j} + 3\hat{k}$,

$\vec{b} = -\beta\hat{i} - \alpha\hat{j} - \hat{k}$ and

$\vec{c} = \hat{i} - 2\hat{j} - \hat{k}$

such that $\vec{a} \cdot \vec{b} = 1$ and $\vec{b} \cdot \vec{c} = -3$, then $\frac{1}{3}\left(\left(\vec{a} \times \vec{b}\right) \cdot \vec{c}\right)$ is equal to _____.

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q88. Numerical · **Mathematics**

The minimum distance between any two points P_1 and P_2 while considering point P_1 on one circle and point P_2 on the other circle for the given circles' equations

$x^2 + y^2 - 10x - 10y + 41 = 0$

$x^2 + y^2 - 24x - 10y + 160 = 0$ is _____.

Open solution

Q89. MCQ (Single Correct Answer) · **Physics**

A Carnot's engine working between 400 K and 800 K has a work output of 1200 J per cycle. The amount of heat energy supplied to the engine from the source in each cycle is :

(A) 2400 J (B) 1600 J (C) 1800 J (D) 3200 J

Open solution

Q90. Numerical · **Physics**

For VHF signal broadcasting, _____ km^2 of maximum service area will be covered by an antenna tower of height 30 m, if the receiving antenna is placed at ground. Let radius of the earth be 6400 km. (Round off to the Nearest Integer) (Take π as 3.14)

Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Answer Key

Chemistry

1. **B**	2. **D**	3. **A**	4. **B**	5. **D**
6. **B**	7. **D**	8. **A**	9. **B**	10. **A**
11. **B**	12. **D**	13. **D**	14. **D**	15. **0**
16. **24**	17. **80**	18. **2**	19. **2**	20. **64**
21. **230**	22. **25**			

Mathematics

23. **D**	24. **C**	25. **B**	26. **B**	27. **A**
28. **A**	29. **B**	30. **D**	31. **D**	32. **C**
33. **B**	34. **C**	35. **C**	36. **B**	37. **4**
38. **6**	39. **904**	40. **1**	41. **481**	42. **6**

Physics

43. **C**	44. **C**	45. **B**	46. **A**	47. **A**
48. **D**	49. **D**	50. **C**	51. **A**	52. **A**
53. **B**	54. **C**	55. **C**	56. **C**	57. **B**
58. **A**	59. **C**	60. **B**	61. **B**	62. **64**
63. **728**	64. **864**	65. **4**	66. **2**	67. **2**
68. **4**	69. **21**	70. **25**		

Out of syllabus

71. **B**	72. **C**	73. **D**	74. **B**	75. **D**
76. **A**	77. **4**	78. **150**	79. **D**	80. **C**
81. **A**	82. **B**	83. **D**	84. **A**	85. **4**
86. **16**	87. **2**	88. **1**	89. **A**	90. **1206**

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

ExamGOAL
PYQ PDF

Your Exam. Our Responsibility.

From first chapter to exam day.

Practise with the questions your exam has already asked, then test yourself under real exam conditions.

PAST YEAR QUESTIONS

Every past paper, chapter-wise.

Previous year questions sorted the way you study, each with a full solution.

- Chapter-wise and year-wise
- Weightage and topic tags on every chapter
- Your progress on each one, at a glance

MOCK TESTS

Mock tests with a real exam feel.

Practise under the same pressure as the real paper, on your phone or on the big screen.

- Timed sections, just like exam day
- Mark questions for review and come back

ANALYSIS

Know your score. Know your rank.

A full report after every test, so you know exactly where you stand.

- Your percentile among everyone who took it
- Predicted real-exam percentile and All India Rank
- Subject breakdown and time analysis

PRACTICE

Solve. Check. Understand.

Learn from every question, not just the ones you get right.

- Instant answers as you practise
- Step-by-step explanations
- See how everyone else answered

NCERT Exemplar
With full solutions

Exam syllabus
Subject and chapter-wise

Bookmarks
Organised in groups

Your notes
On any question

ExamGOAL
LEARNING APP

ExamGOAL: Exam Preparation App

One app for every exam. Free trial for new users, on Android and the web.

GET IT ON Google Play

Scan to get the app

www.examgoal.com
Practice and mock tests

questions.examside.com
Free solutions

pdf.questions.examgoal.com
Free PYQ PDFs